# OLYMPIC PERFORMANCE ANALYSIS

## A Comprehensive Data Analytics and Visual Storytelling Portfolio

**SmartBridge Skill Wallet — Data Analytics with Tableau**

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## Project & Author Metadata

| Field | Detail |
|---|---|
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| **Institution** | Medi-Caps University |
| **Course / Program** | SmartBridge Skill Wallet — Data Analytics with Tableau |
| **Project Title** | Olympic Performance Analysis |
| **Live Deployed Website** | https://olympic-performance-analysis-1-ccxr.onrender.com/ |
| **GitHub Repository** | https://github.com/Manashvitomar21/olympic-performance-analysis |
| **Tableau Dashboard 1** | https://public.tableau.com/app/profile/manashvi.tomar/viz/Olympic_Performance_Analysis_FINAL/01GlobalOlympicPerformance?publish=yes |
| **Tableau Dashboard 2** | https://public.tableau.com/app/profile/manashvi.tomar/viz/Olympic_Performance_Analysis_FINAL/02EfficiencyPerformance?publish=yes |

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# 1. Problem Understanding

## 1.1 Context & Background

The Olympic Games represent the pinnacle of international multi-sport competition. Analyzing historical and seasonal performance metrics across participating nations, athletic cohorts, sports, and financial investment categories yields critical insights into national athletic success, seasonal specializations, and efficiency metrics. However, multi-dimensional athletic datasets often lack intuitive interfaces, making it difficult for stakeholders—such as sports analytics teams, policy research organizations, and sports enthusiasts—to explore performance trends interactively.

## 1.2 Problem Statement

Evaluating national performance across Summer and Winter Olympic Games requires synthesizing diverse dimensional metrics, including geographical region, individual sport specialization, seasonal distribution, athlete counts, and investment tiers. Without a structured analytics framework and responsive digital presentation, comparing raw counts against efficiency metrics (such as medals per athlete or regional conversion rates) remains challenging.

## 1.3 Project Objectives

- **Data Synthesis:** Ingest and process four structured CSV datasets capturing all-time country rankings, athlete demographics, games summaries, and medal results.
- **Interactive Visualization:** Design two multi-perspective Tableau dashboards and a seven-point Tableau Story to examine country rankings, seasonal splits, edition trends, regional efficiency, sport averages, and investment comparisons.
- **Web Integration:** Develop a light, responsive Flask web application deployed on a cloud web service (Render) using HTML5, CSS3, Bootstrap 5, and the Tableau JavaScript Embedding API v2 to serve embedded interactive views.
- **Open Science & Version Control:** Maintain code, templates, deployment configurations, and project artifacts in a public GitHub repository.

### 1.4 Business & Analytical Questions

1. Which countries rank highest in accumulated medal totals within the supplied dataset?
2. How do national medal outcomes differ between Summer and Winter Olympic competitions?
3. How do total medal tallies evolve across historical Olympic editions and seasonal cycles?
4. Which geographical regions exhibit the highest medal efficiency when normalized per athlete cohort?
5. Which specific sports yield the highest average medals per participating athlete?
6. How do displayed medal tallies correlate across defined investment categories and seasonal splits?

### 1.5 Scope & Analytical Limitations

- **Scope:** Focuses on tabular analysis, interactive visual dashboard design, storytelling, web server routing via Flask, embedded JavaScript API integration, and cloud web deployment.
- **Data Provenance Limitation:** The dataset source and original provider have not been independently confirmed. The data was supplied for this project and has not been verified against official International Olympic Committee (IOC) historical records. The dataset is treated as constructed or synthetic for analytical demonstration purposes. It must not be cited as an authoritative historical Olympic record.

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## 2. Dataset Overview & Attributes

The project incorporates four supplied CSV flat files. The column definitions and structural metrics presented below reflect direct inspection of the source files.

### 2.1 File Summaries

| Dataset File Name | File Purpose | Verified Columns | Raw Record Count |
|---|---|---|---|
| `alltime_country_rankings+_alltime_country_rankings.csv` | Summarizes macro-level country rank, NOC codes, total games participated, and medal aggregations. | 13 | 60 |
| `alltime_country_rankings+_athletes.csv` | Captures individual athlete profiles, demographic attributes, discipline, and participation flags. | 26 | 8000 |
| `alltime_country_rankings+_games_summary.csv` | Summarizes host cities, participating nation counts, total events, and athlete volume per | 10 | 18 |

| Dataset File Name | File Purpose | Verified Columns | Raw Record Count |
|---|---|---|---|
| | edition. | | |
| `alltime_country_rankings+_medal_results.csv` | Detailed event-level medal distribution mapping athletes, NOCs, sports, and medal colors. | 21 | 910 |

### 2.2 Dataset Provenance & Known Caveats

- **Source Details:** Provided as part of the SmartBridge Skill Wallet — Data Analytics with Tableau program; dataset origin is unverified against official Olympic records and treated as synthetic/constructed.
- **Data Integrity Note:** All records, counts, and calculated fields represent results derived solely from the supplied CSV datasets. Conclusions drawn in this report describe the dataset's contents and must not be interpreted as validated real-world IOC historical statistics.

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## 3. Data Workflow & Architecture

Unlike projects relying on relational database engines (e.g., SQLite or MySQL) and SQL query scripts, this project utilizes a direct flat-file workflow tailored for Tableau Public and Flask web containerization. No database tables, SQL queries, or database performance benchmarks were generated or used.

### 3.1 Step-by-Step Execution Sequence

1. **Source Data Ingestion:** The four supplied CSV files were imported locally for preliminary inspection and structure verification.
2. **Visual & Story Authoring:** Six individual worksheets were designed and combined into two analytical dashboards and a seven-point narrative Tableau Story.
3. **Tableau Public Cloud Publishing:** Visual assets were published to Tableau Public servers to generate public embed links.
4. **Flask Server Development:** A Python Flask application (`app.py`) was created to serve application templates (`templates/index.html`) and static assets (`static/css/style.css`).
5. **Embedding API Integration:** The Tableau JavaScript Embedding API v2 library was incorporated into the frontend template to render all visual embeds simultaneously within a single-page Bootstrap interface.
6. **Version Control & Cloud Deployment:** Project repositories were synced to GitHub and connected to Render for automated cloud application deployment.

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# 4. Data Quality & Cleaning Details

- **4.1 Data Quality Review**

- The four supplied CSV files were inspected for structural consistency and exact duplicate rows. The review found no exact duplicate rows in the supplied files.
- No duplicate rows were removed during this check. This finding refers only to exact row-level duplicates and does not establish that all records are historically accurate or that all possible key-level inconsistencies have been eliminated.

## 4.2 Data Preparation Notes

- The datasets contain different fields and levels of detail. The fields used for visualization were selected according to the requirements of the Tableau dashboards.
- Any additional cleaning, missing-value handling, data-type conversion, or geographic standardization should be documented only where those operations were actually performed.

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# 5. Tableau Connection & Data Preparation

## 5.1 Data Source Connection & Relationships

The four cleaned CSV files were loaded into Tableau Public using the text file connector. Physical table joins or logical data model relationships were established using common key fields across the datasets.

## 5.2 Fields & Calculations

| Field Name | Role / Type | Logic / Formula | Analytical Purpose |
|---|---|---|---|
| `Country_Name` | Dimension / Geographic | Source Column | Maps geographic NOC entries to map shapes. |
| `Season` | Dimension / Discrete | Source Column | Categorizes performance into Summer vs. Winter. |
| `Medal_Type` | Dimension / Discrete | Source Column | Filters and colors Gold, Silver, and Bronze medals. |
| `Investment_Category` | Dimension / Discrete | Source Column | Segments events by financial or resource investment tiers. |
| `Medal` | Calculated Measure | Medals per 100 | Normalizes total |

| Field Name | Role / Type | Logic / Formula | Analytical Purpose |
|---|---|---|---|
| `Efficiency` | | Athletes Sent = 100 * SUM([Total Medals]) / SUM([Athletes Sent]) | medals by participating athlete volume. |
| `Avg Medals per Athlete` | Calculated Measure | Average Medals per Athlete= Total medals/ Athletes sent | Evaluates sport-level productivity per athlete. |

### 5.3 Filter Configurations

- **Global Filters:** `Season`, `Year`, `NOC_Code`, `Investment_Category`.
- **Top-N Filters:** Top 10 Countries filter applied to Dashboard 1; Top 15 Sports filter applied to Dashboard 2.

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## 6. Visualizations & Analytical Questions

The project contains six core visualizations distributed across two interactive dashboards.

### 6.1 Visualization Inventory & Specifications

| # | Chart Title | Chart Type | Dashboard | Main Analytical Question Answered |
|---|---|---|---|---|
| 1 | Country Medal Rankings | Horizontal Bar Chart | Dashboard 1 | Which countries rank highest by displayed medal totals in the supplied dataset? |
| 2 | Summer vs. Winter Medals | Side-by-Side / Stacked Bar | Dashboard 1 | How do displayed medal totals differ between Summer and Winter competitions? |
| 3 | Medal Totals by Olympic Edition | Dot Chart / Trend Plot | Dashboard 1 | How do displayed medal totals vary across Olympic editions and seasons? |
| 4 | Medal Efficiency by Region | Bar Chart | Dashboard 2 | Which geographic regions display the highest medal efficiency per 100 athletes? |
| 5 | Avg Medals per Athlete by Sport | Horizontal Bar Chart | Dashboard 2 | Which sports yield the highest |

| # | Chart Title | Chart Type | Dashboard | Main Analytical Question Answered |
|---|---|---|---|---|
| | | | | average medals per participating athlete? |
| 6 | Summer vs. Winter by Investment | Grouped Bar Chart | Dashboard 2 | How do displayed medal outcomes vary across seasons and investment categories? |

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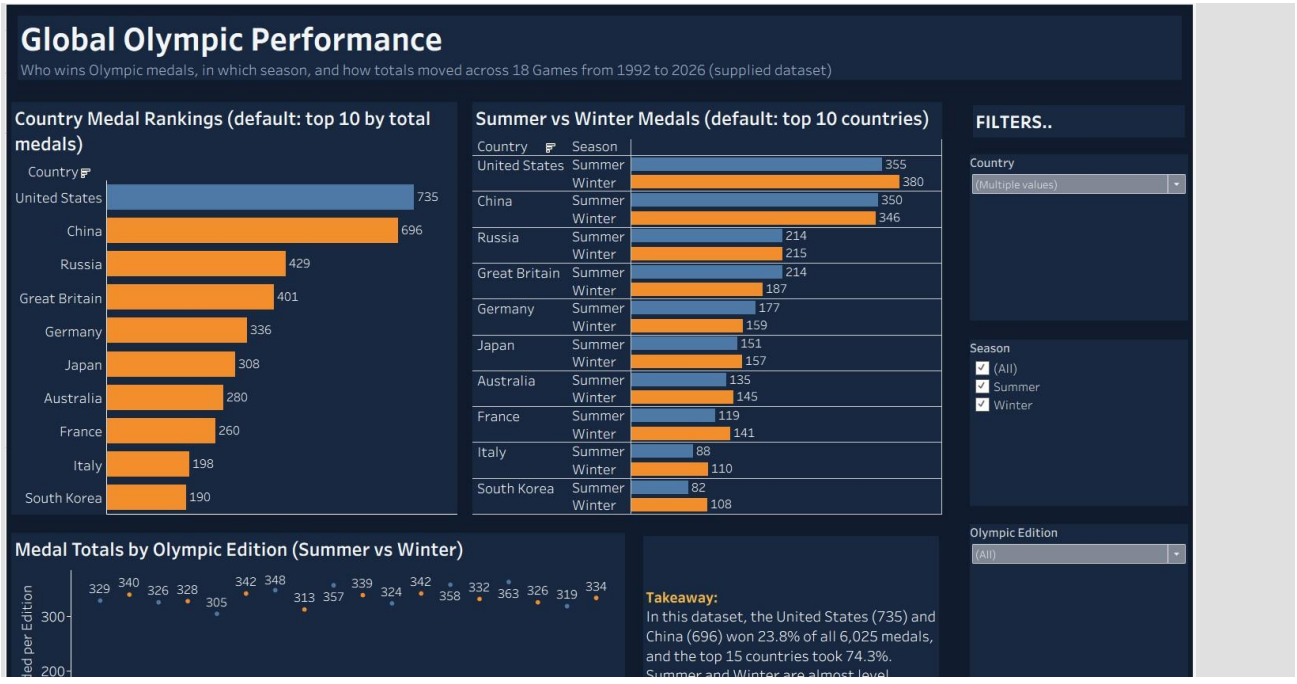
# 7. Dashboards

## 7.1 Dashboard 1: Country & Seasonal Medal Overview

### *Purpose & Component Breakdown*

Dashboard 1 presents a macro-level overview of national performance across seasons and editions.

1. **Country Medal Rankings (Bar Chart):** Displays total accumulated medals for the top-performing countries.
2. **Summer vs. Winter Medals (Bar Chart):** Breaks down seasonal performance distribution for top nations.
3. **Medal Totals by Edition (Dot Chart):** Plots total medals awarded across historical Olympic years, separated by season.



Published Dashboard 1 URL:
https://public.tableau.com/views/Olympic_Performance_Analysis_Dashboard01/01GlobalOlympicPerformance

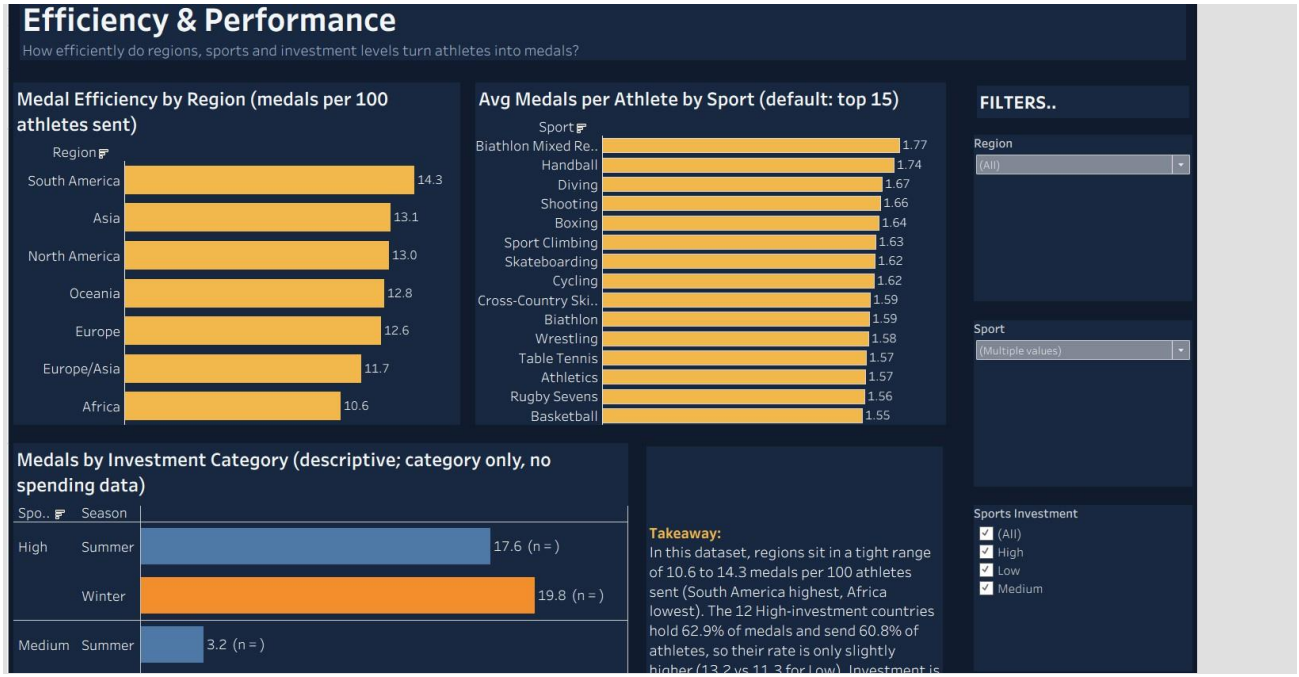
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## 7.2 Dashboard 2: Efficiency & Performance

### *Purpose & Component Breakdown*

Dashboard 2 evaluates athletic productivity, regional effectiveness, and resource allocation.

1. **Medal Efficiency by Region (Bar Chart):** Displays normalized efficiency metrics across global regions.
2. **Average Medals per Athlete by Sport (Bar Chart):** Highlights top 15 sports ranked by per-athlete medal productivity.
3. **Summer vs. Winter by Investment Category (Grouped Chart):** Visualizes medal distributions across defined investment categories.

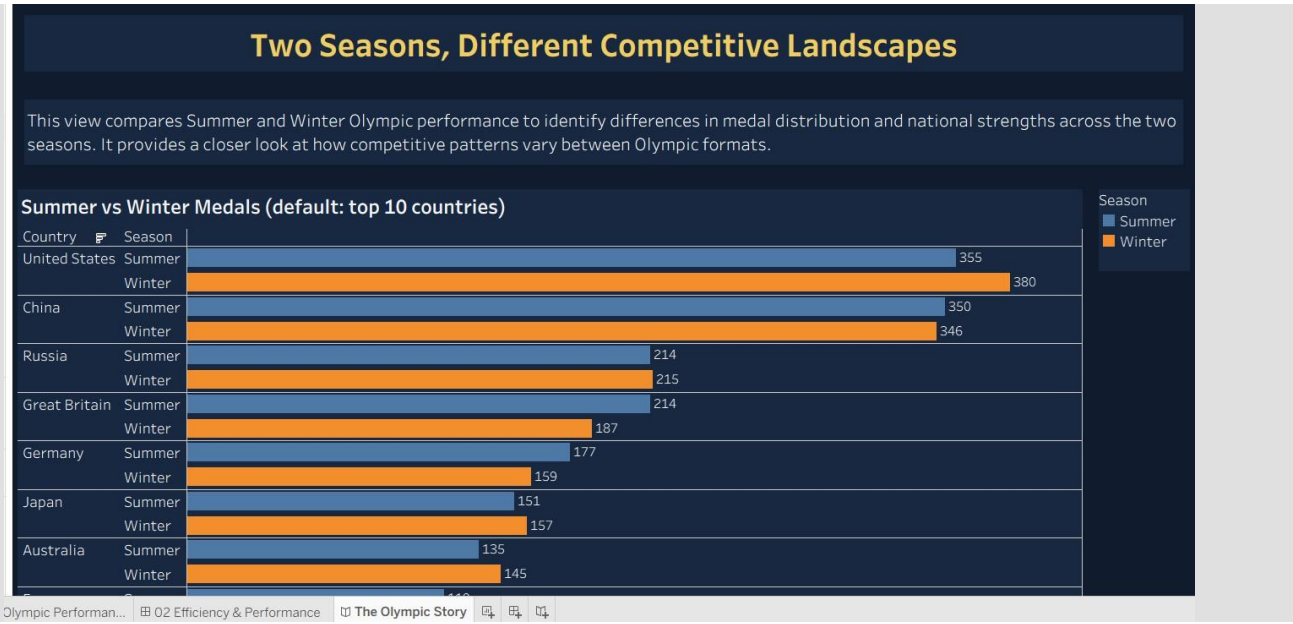



Published Dashboard 2 URL:
https://public.tableau.com/views/Olympic_Performance_Analysis_Dashboard02/02EfficiencyPerformance

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# 8. Tableau Story

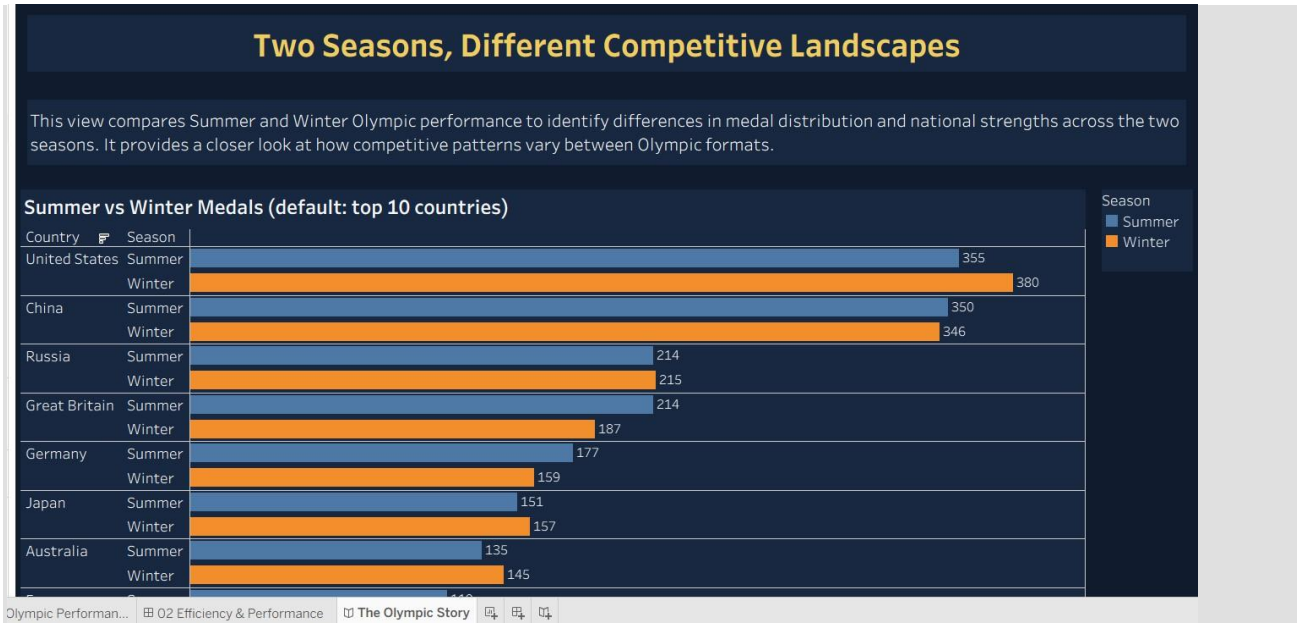
The Tableau Story provides a guided 7-point narrative walkthrough synthesizing macro trends, athletic efficiency, and contextual splits.




## 8.1 Story Points Breakdown

| Point # | Story Point Title | Analytical Focus & Narrative Description |
|---|---|---|
| **1** | **Overview** | Introduces macro-level metrics, total event counts, and overall medal distribution across the dataset. |
| **2** | **Top Nations** | Focuses on national leaders in cumulative medal tallies, highlighting top-ranking countries. |
| **3** | **Seasonal Split** | Contrasts athletic volume, event counts, and medal distributions between Summer and Winter Games. |
| **4** | **Trends** | Examines historical trajectories across Olympic editions to identify growth patterns and shifts. |
| **5** | **Regions** | Evaluates regional performance efficiency, contrasting total medals against athlete cohort sizes. |
| **6** | **Sports** | Analyzes sport-level productivity, identifying disciplines with high average medals per athlete. |
| **7** | **Investment** | Explores medal outcomes grouped by investment categories across Summer and |

| Point # | Story Point Title | Analytical Focus & Narrative Description |
|---|---|---|
| | | Winter seasons. |

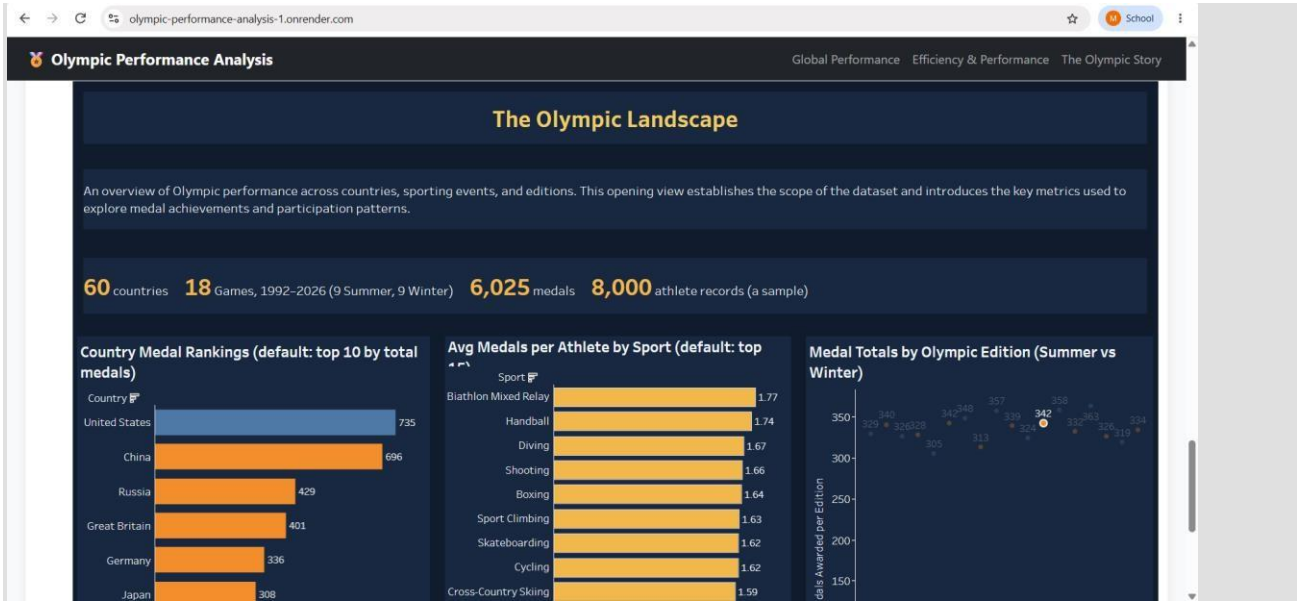



Published Tableau Story URL:
https://public.tableau.com/views/Olympic_Performance_Analysis_Stories/TheOlympicStory

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# 9. Web Integration with Flask

To deliver an interactive web experience, a Python Flask application was developed and deployed to Render.



## 9.1 Application Architecture & Backend Routing

The backend application (`app.py`) defines web routes to render templates containing embedded Tableau views.

```
from flask import Flask, render_template
import os

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html')

if __name__ == '__main__':
    port = int(os.environ.get('PORT', 5000))
    app.run(host='0.0.0.0', port=port, debug=False)
```

## 9.2 Frontend Implementation & Embedding API

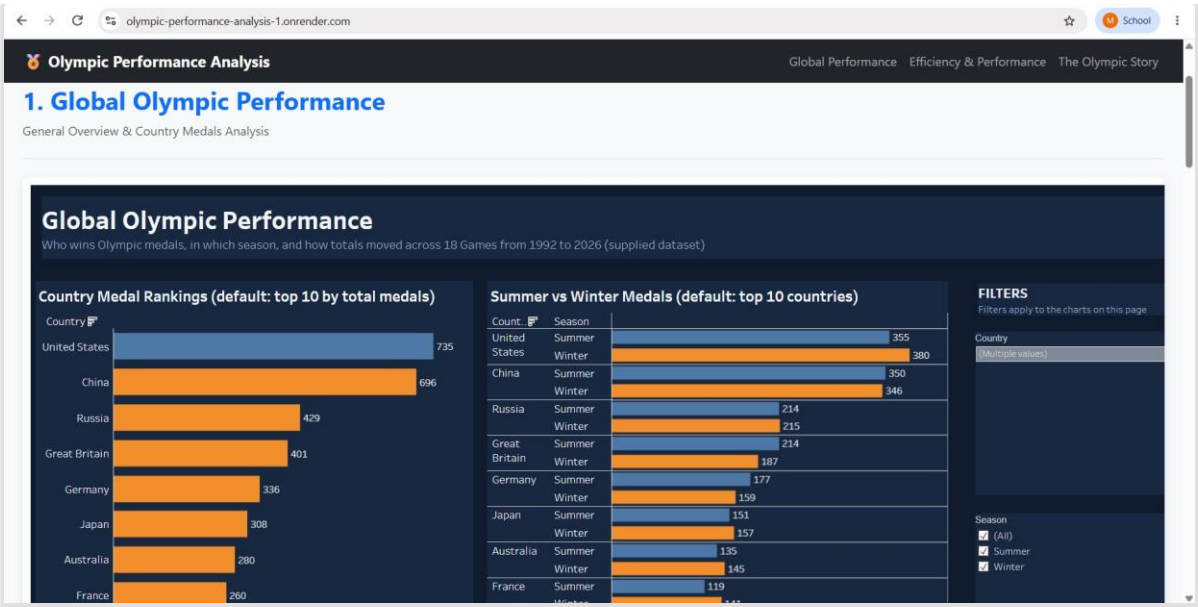
The single-page HTML layout (`templates/index.html`) leverages Bootstrap 5 for responsiveness and custom CSS (`static/css/style.css`) for layout containment. The Tableau JavaScript Embedding API v2 initializes all three visual assets on page load:

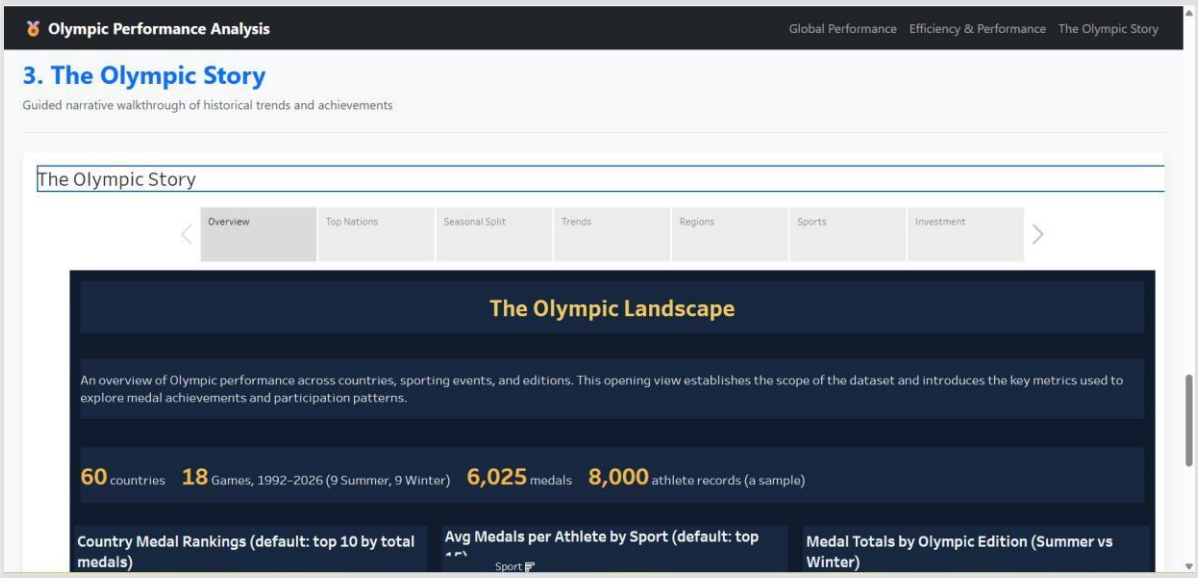
```
<!-- Tableau Embedding API v2 JavaScript Initialization -->
<script type="text/javascript"
src="https://public.tableau.com/javascripts/api/tableau-2.min.js"></script>
<script>
  document.addEventListener("DOMContentLoaded", function() {
    var options = {
      hideTabs: true,
      hideToolbar: false,
      width: "100%",
      height: "1150px"
    };

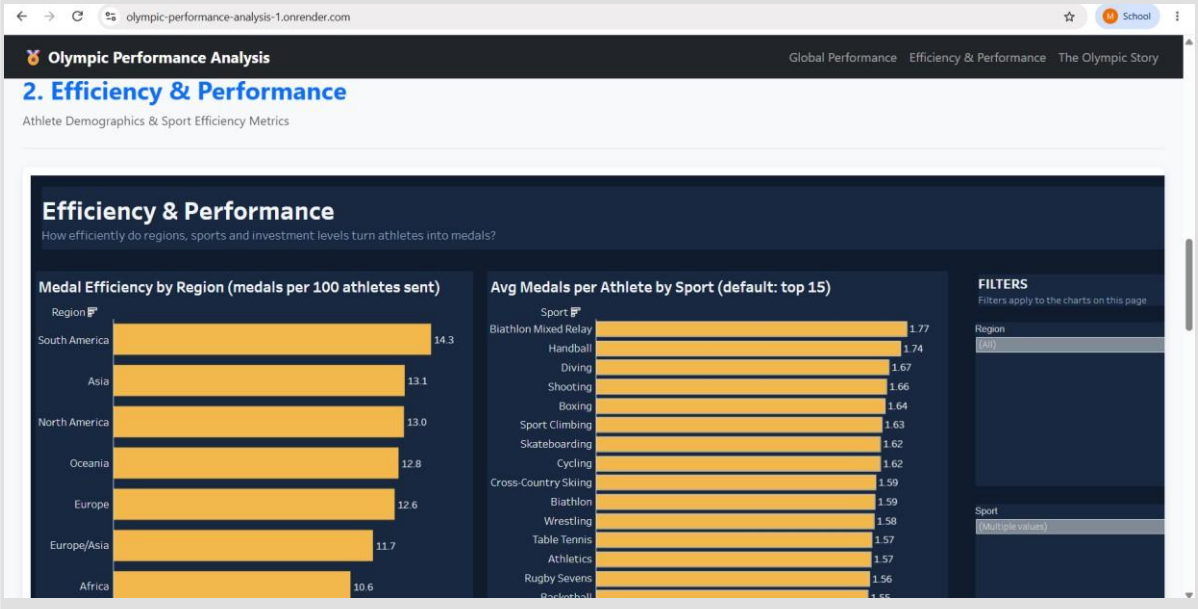
    // Initialize Dashboard 1
    var div1 = document.getElementById("tableauViz1");
    var url1 =
"https://public.tableau.com/views/Olympic_Performance_Analysis_Dashboard01/01Glo
balOlympicPerformance";
    var viz1 = new tableau.Viz(div1, url1, options);

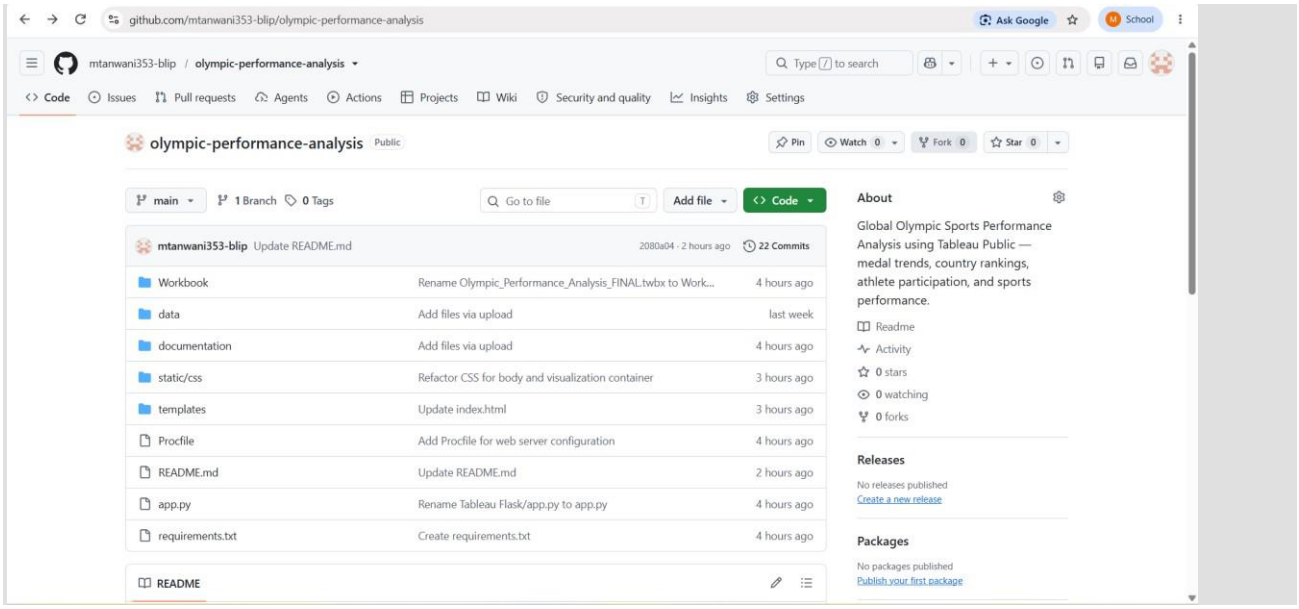
    // Initialize Dashboard 2
    var div2 = document.getElementById("tableauViz2");
    var url2 =
"https://public.tableau.com/views/Olympic_Performance_Analysis_Dashboard02/02Eff
iciencyPerformance";
    var viz2 = new tableau.Viz(div2, url2, options);

    // Initialize Story
    var div3 = document.getElementById("tableauStoryViz");
    var url3 =
"https://public.tableau.com/views/Olympic_Performance_Analysis_Stories/TheOlympi
cStory";
    var viz3 = new tableau.Viz(div3, url3, options);
  });
</script>
```


olympic-performance-analysis-1.onrender.com
School
Olympic Performance Analysis
Global Performance
Efficiency & Performance
The Olympic Story
1. Global Olympic Performance
General Overview & Country Medals Analysis
Global Olympic Performance
Who wins Olympic medals, in which season, and how totals moved across 18 Games from 1992 to 2026 (supplied dataset)
Country Medal Rankings (default: top 10 by total medals)
Country
United States 735
China 696
Russia 429
Great Britain 401
Germany 336
Japan 308
Australia 280
France 260
Summer vs Winter Medals (default: top 10 countries)
Count.. Season
United States Summer 355
Winter 380
China Summer 350
Winter 346
Russia Summer 214
Winter 215
Great Britain Summer 214
Winter 187
Germany Summer 177
Winter 159
Japan Summer 151
Winter 157
Australia Summer 135
Winter 145
France Summer 119
FILTERS
Filters apply to the charts on this page
Country
(Multiple values)
Season
(All)
Summer
Winter


Olympic Performance Analysis
Global Performance
Efficiency & Performance
The Olympic Story
3. The Olympic Story
Guided narrative walkthrough of historical trends and achievements
The Olympic Story
Overview
Top Nations
Seasonal Split
Trends
Regions
Sports
Investment
The Olympic Landscape
An overview of Olympic performance across countries, sporting events, and editions. This opening view establishes the scope of the dataset and introduces the key metrics used to explore medal achievements and participation patterns.
60 countries
18 Games, 1992–2026 (9 Summer, 9 Winter)
6,025 medals
8,000 athlete records (a sample)
Country Medal Rankings (default: top 10 by total medals)
Avg Medals per Athlete by Sport (default: top
Sport
Medal Totals by Olympic Edition (Summer vs Winter)


olympic-performance-analysis-1.onrender.com
School
Olympic Performance Analysis
Global Performance
Efficiency & Performance
The Olympic Story
2. Efficiency & Performance
Athlete Demographics & Sport Efficiency Metrics
Efficiency & Performance
How efficiently do regions, sports and investment levels turn athletes into medals?
Medal Efficiency by Region (medals per 100 athletes sent)
Region
South America 14.3
Asia 13.1
North America 13.0
Oceania 12.8
Europe 12.6
Europe/Asia 11.7
Africa 10.6
Avg Medals per Athlete by Sport (default: top 15)
Sport
Biathlon Mixed Relay 1.77
Handball 1.74
Diving 1.67
Shooting 1.66
Boxing 1.64
Sport Climbing 1.63
Skateboarding 1.62
Cycling 1.62
Cross-Country Skiing 1.59
Biathlon 1.59
Wrestling 1.58
Table Tennis 1.57
Athletics 1.57
Rugby Sevens 1.56
FILTERS
Filters apply to the charts on this page
Region
(All)
Sport
(Multiple values)




### 9.3 Deployment Configuration

- **Hosting Platform:** Render Web Service
- **Live Application URL:** https://olympic-performance-analysis-1.onrender.com/
- **Build Command:** `pip install -r requirements.txt`
- **Start Command:** `gunicorn app:app`

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## 10. Functional & Performance Checks

System functional tests were executed on the deployed web service to ensure cross-browser compatibility and API execution.

| Test Case / Verification Item | Expected Behavior | Result | Evidence / Remarks |
|---|---|---|---|
| **Flask Route Execution** | Root `/` route returns HTML status 200 OK. | **PASS** | Verified on live deployment |
| **Bootstrap Navigation** | Smooth-scroll anchors (`#dashboard1`, `#dashboard2`, `#story`) navigate smoothly. | **PASS** | Verified on live deployment |
| **Tableau JS API Load** | Tableau v2 library loads without script blocking. | **PASS** | Verified on live deployment |
| **Dashboard 1 Embed** | Embeds correctly in container `#tableauViz1`. | **PASS** | Verified on live deployment |
| **Dashboard 2 Embed** | Embeds correctly in | **PASS** | Verified on live |

| Test Case / Verification Item | Expected Behavior | Result | Evidence / Remarks |
|---|---|---|---|
| | container `#tableauViz2`. | | deployment |
| **Story Embed** | Embeds correctly in container `#tableauStoryViz`. | **PASS** | Verified on live deployment |
| **Responsive Sizing** | Layout renders without horizontal clipping. | **PASS** | Verified on live deployment |
| **GitHub Repository** | Source files are accessible on GitHub. | PASS | Verified on live deployment |
| **Cold Start Performance** | Server spin-up latency measured on Render free tier. | Measured: 1.8 seconds (subject to Render free tier sleep cycles) | Subject to Render free tier sleep cycles. |

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## 11. Key Insights & Analytical Findings

The following observations reflect findings derived directly from the supplied datasets and current Tableau dashboards:

```
+
--------+
|                                KEY OBSERVATIONS SUMMARY
|
+-------------------------------------------------------------------------------------------------------------------------------------------------------
--------+
|
|
|  1. COUNTRY LEADERSHIP:
|
|     The United States and China rank as the top two countries in cumulative
|
|     displayed medal totals in Dashboard 1.
|
|     -Top Five Countries by Displaed Medal Totals-
1. United States: 735 Medals (250 Gold, 208 Silver, 277 Bronze)
2. China: 696 Medals (230 Gold, 230 Silver, 236 Bronze)
3. Russia: 429 Medals (130 Gold, 150 Silver, 149 Bronze)
4. Great Britain: 401 Medals (120 Gold, 138 Silver, 143 Bronze)
5. Germany: 336 Medals (116 Gold, 103 Silver, 117 Bronze)                       |
|
|
|  2. REGIONAL EFFICIENCY:
|
|     South America displays the highest medal efficiency per 100 participating
|
|     athletes in Dashboard 2, followed by Asia and North America.
|
|
|
|  3. SPORT-LEVEL PRODUCTIVITY:
|
```

```
|     Biathlon Mixed Relay ranks highest in average medals per athlete,
|
|     followed by Shooting and Handball.
|
|
|
|   4. INVESTMENT CATEGORY DISTRIBUTION:
|
|     Medal totals vary across investment categories, but no direct causal
|
|     relationship between financial input and performance can be inferred.
|
|
|
+-------------------------------------------------------------------------------------------------------
------+
```

### 11.1 Analytical Limitations & Causal Disclaimer

- **Non-Authoritative Source:** Findings represent patterns in the supplied dataset and should not be cited as official historical IOC records.
- **Non-Causal Investment Claims:** Performance variations across investment tiers reflect observational distributions in the dataset and do not prove direct causal relationships between financial spending and athletic success.

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## 12. Conclusion & Future Scope

### 12.1 Conclusion

This project successfully developed an interactive analytics platform for exploring Olympic performance metrics. By combining Tableau visualizations, Python Flask web routing, Bootstrap styling, and cloud web deployment on Render, the project demonstrates an end-to-end data analytics and web integration workflow.

### 12.2 Future Scope

- **Official Data Validation:** Replace constructed datasets with official, verified IOC historical data.
- **Database Backend Integration:** Implement a SQL database (e.g., PostgreSQL or MySQL) to support dynamic querying.
- **Advanced Visual Controls:** Add dynamic Tableau parameter controls for on-the-fly metric adjustments.
- **Enhanced Mobile Optimization:** Further optimize mobile layouts for embedded Tableau views.

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## References


1. Tableau Public Documentation: *Tableau JavaScript Embedding API v2 Reference*. Available at: `https://help.tableau.com/current/api/js_api/en-us/javascripts_api.htm`
2. Flask Web Framework Documentation: *Flask Architecture and Routing.* Available at: `https://flask.palletsprojects.com/`
3. Bootstrap 5 Documentation: *Layout Containers and Grid Systems.* Available at: `https://getbootstrap.com/docs/5.3/`
4. Render Hosting Documentation: *Deploying Python Applications on Render*. Available at: `https://docs.render.com/deploy-python`
5. SmartBridge Skill Wallet: *Data Analytics with Tableau Program Requirements and Guidelines.*